

HW 0519

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6.

(a)

The OLS estimates for 1987 and 1988 are highly consistent and statistically significant across both years, with very stable coefficients such as the return on experience (0.1270 vs. 0.1265). In these individual-year cross-sectional estimations, you are assuming parameter homogeneity, meaning that the regression coefficients are identical for all individuals within that year, so every person faces the exact same marginal effects. Furthermore, you assume the absence of correlated unobserved heterogeneity, meaning any individual-specific, time-invariant traits (like innate ability) are assumed to be uncorrelated with the explanatory variables, treating them as part of the purely random error term.

(b)

The critical difference lies in the assumptions about these components: the panel model recognizes individual-specific, time-invariant effects (u_i)—such as innate ability or motivation—that stay constant over time for each individual, whereas the individual-year OLS models pool these effects into a single error term and strictly assume they are uncorrelated with the regressors. Furthermore, while the OLS models assume that the idiosyncratic errors are independent across all observations, the panel structure allows for the fact that observations for the same individual across different years (1987 and 1988) are likely to be serially correlated due to the persistent presence of u_i .

(c)

The most substantial differences are experience (*EXPER*) and living in the south (*SOUTH*). The coefficient for *EXPER* drops sharply from approximately 0.127 in the OLS models to 0.0575 in the FE model, indicating that OLS suffered from severe upward omitted variable bias by mistakenly attributing the positive effects of unobserved traits—like innate ability or drive (u_i)—to job experience. Conversely, the negative impact of living in the south (*SOUTH*) becomes considerably stronger, shifting from around -0.21 / -0.23 in the OLS estimations to -0.3261 under the FE model, which demonstrates how controlling for time-invariant individual heterogeneity fundamentally alters the estimated behavioral relationships.

(d)

The degrees of freedom for the F -distribution under the null hypothesis of no individual differences are (715, 712). The numerator degrees of freedom are calculated as $df_1 = N - 1 = 716 - 1 = 715$, representing the number of individual restrictions, while

the denominator degrees of freedom are $df_2 = NT - N - K = 1432 - 716 - 4 = 712$, where $N = 716$, $T = 2$, and $K = 4$ (the number of time-varying regressors: *EXPER*, *EXPER*², *SOUTH*, *UNION*). Given these exceptionally large degrees of freedom, the 1% level of significance critical value from the F -distribution table is approximately 1.15. Since the calculated F -statistic of 11.68 is vastly larger than this critical value, we strongly reject the null hypothesis and conclude that individual-specific differences (u_i) are highly significant, meaning a pooled OLS model is statistically inappropriate.

(e)

When estimating without cluster-robust standard errors (the standard FE in column 3), you assume that the idiosyncratic errors (e_{it}) are independent and identically distributed (i.i.d.) across both individuals and time, meaning there is no heteroskedasticity and zero serial correlation within the same person's observations. By contrast, when estimating with cluster-robust standard errors (column 4), you relax this rigid assumption by allowing the errors (e_{it}) for any given individual to have arbitrary heteroskedasticity and serial correlation over time (between 1987 and 1988), while still assuming independence across different individuals.

Comparing the two columns, the cluster-robust standard errors in column (4) are generally larger than the conventional ones in column (3). The most substantially different standard error belongs to the *SOUTH* variable, which nearly doubles from 0.1258 (non-robust) to 0.2495 (robust), rendering the coefficient statistically insignificant. The standard error for *UNION* also increases from 0.0312 to 0.0367, demonstrating that ignoring the positive serial correlation within individual "clusters" leads conventional FE to overstate the precision of the estimates and inflate t -statistics.

(f)

Apart from the intercepts, the coefficients that show the most substantial differences between the random effects (RE) estimates in column (5) and the fixed effects (FE) estimates in column (3) are experience (*EXPER*) and living in the south (*SOUTH*). The coefficient for *EXPER* is 0.0575 in FE but increases to 0.0986 in RE, while the coefficient for *SOUTH* changes from -0.3261 in FE to -0.2326 in RE.

To perform the Hausman test, we must use the traditional, non-robust FE estimates from column (3) rather than the cluster-robust ones in column (4). This is because the standard Hausman test statistic formula, $H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})'[Var(\hat{\beta}_{FE}) - Var(\hat{\beta}_{RE})]^{-1}(\hat{\beta}_{FE} - \hat{\beta}_{RE})$, strictly relies on the theoretical assumption that the RE estimator is fully efficient under the null hypothesis (H_0). This efficiency property implies that $Cov(\hat{\beta}_{FE} - \hat{\beta}_{RE}, \hat{\beta}_{RE}) = 0$, allowing the variance of the difference to simplify beautifully to $Var(\hat{\beta}_{FE}) - Var(\hat{\beta}_{RE})$. If we were to use the cluster-robust errors from column (4), this mathematical simplification breaks down, making the standard test statistic formula invalid.

Based on the substantial differences between the FE and RE coefficients for key variables like *EXPER* and *SOUTH*, a formal Hausman test would yield a highly significant statistic, leading to a strong rejection of the null hypothesis that the unobserved individual effects (u_i) are uncorrelated with the regressors. Because the null hypothesis is rejected, the RE estimator is biased and inconsistent due to endogeneity. Therefore, we

conclude that random effects estimation is not appropriate for this model, and the fixed effects approach should be utilized instead.

20.

(a)

Table 1: (a)

	<i>Dependent variable:</i>
	readscore
small	5.823*** (0.989)
aide	0.818 (0.953)
tchexper	0.492*** (0.070)
boy	-6.156*** (0.796)
white_asian	3.906*** (0.954)
freelunch	-14.771*** (0.890)
Constant	437.764*** (1.346)
Observations	5,766
R ²	0.097
Adjusted R ²	0.096
Residual Std. Error	30.186 (df = 5759)
F Statistic	102.932*** (df = 6; 5759)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Based on the Pooled OLS regression results, students in small classes perform significantly better in reading, with an estimated score increase of 5.823 points ($p < 0.01$) compared to the reference group, whereas having a teacher's aide does not significantly improve reading scores ($\hat{\beta} = 0.818$, $p > 0.1$). Teacher's experience has a statistically significant positive impact, where each additional year of teaching experience is associated with a 0.492-point increase ($p < 0.01$) in student reading scores. Both student sex and race make a highly significant difference: boys score an average of 6.156 points lower than girls ($p < 0.01$), while white or Asian students score an average of 3.906 points higher

than students of other racial backgrounds ($p < 0.01$), holding other socioeconomic and classroom factors constant.

(b)

Table 2: (b)	
	<i>Dependent variable:</i>
	readscore
small	6.490*** (0.913)
aide	0.996 (0.882)
tchexper	0.286*** (0.071)
boy	−5.456*** (0.728)
white_asian	8.028*** (1.536)
freelunch	−14.594*** (0.880)
Observations	5,766
R ²	0.078
Adjusted R ²	0.064
F Statistic	79.647*** (df = 6; 5681)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Overall, the core qualitative conclusions do not change because the main policy variables—small remains a highly significant positive driver of reading scores (6.490, $p < 0.01$), aide stays statistically insignificant (0.996, $p > 0.1$), and tchexper, boy, and freelunch retain their original signs and high statistical significance ($p < 0.01$). However, controlling for school fixed effects reveals crucial sorting biases: the return to teacher experience decreases from 0.492 to 0.286, indicating that more experienced teachers are disproportionately employed by higher-performing schools, while the coefficient for white-asian more than doubles from 3.906 to 8.028 ($p < 0.01$). This substantial shift in the race coefficient demonstrates that the Pooled OLS estimate was heavily confounded by across-school sorting (e.g., minority students clustering in schools with lower unobserved baseline qualities), proving that the racial achievement gap is actually much wider when compared within the exact same school environment.

(c)

Table 3: F-Test for Significance of School Fixed Effects

Test Type	F-Statistic	df1	df2	p-value
F-Test for Individual Effects	16.698	78	5,681	$< 2.2 \times 10^{-16}$ ***

Note: *** indicates significance at the 1% level.

H_0 : No school fixed effects ($\alpha_i = 0$ for all i).

The highly significant F -test ($F = 16.698$, $p < 2.2 \times 10^{-16}$) strongly rejects the null hypothesis of poolability, confirming that substantial, unobserved school-level heterogeneity (α_i) exists within the dataset. In econometrics, the inclusion of these significant fixed effects will have little to no influence on the coefficient estimates of the remaining explanatory variables if and only if the independent variables are orthogonal to the unobserved group-specific effects, mathematically expressed as $\text{Cov}(X_{ij}, \alpha_i) = 0$. In the context of the STAR experiment, because students and teachers were randomly assigned within each school, the classroom treatment variables (`small` and `aide`) are guaranteed by the experimental design to be uncorrelated with any unobserved school-level characteristics (such as school funding or principal leadership). Consequently, omitting these school-specific intercepts in the Pooled OLS model does not introduce omitted variable bias for the randomized policy variables, which perfectly explains why the estimated treatment effect of a small class remains remarkably stable, shifting only slightly from 5.823 to 6.490.

(d)

Table 4: Breusch-Pagan Lagrange Multiplier Test for School Random Effects

Null Hypothesis (H_0)	LM Statistic (χ^2)	df	p-value
No random school effects ($\sigma_\alpha^2 = 0$)	6,677.4	1	$< 2.2 \times 10^{-16}$ ***

Note: *** indicates statistical significance at the 1% level.

Alternative Hypothesis (H_1): Significant school random effects exist.

The Breusch-Pagan Lagrange Multiplier test strongly rejects the null hypothesis ($\chi^2 = 6677.4$, $p < 2.2 \times 10^{-16}$), indicating that significant unobserved school-level random effects exist and Pooled OLS is inappropriate. While the classroom assignment variables (`small` and `aide`) remain highly consistent across all specifications due to within-school randomization, demographic variables such as `white_asian` exhibit pronounced variations, dropping from 8.028 in the Fixed Effects model to 5.922 in the Random Effects model. This noticeable shift suggests that school-level random effects (α_i) are likely correlated with student background characteristics like `white_asian` and `freelunch`, as residential and socioeconomic sorting across neighborhoods causes specific racial and income groups to cluster within particular schools, thereby violating the crucial Random Effects assumption ($\text{Cov}(X_{ij}, \alpha_i) = 0$) and rendering the Random Effects estimates biased.

(e)

At the 5% significance level, the individual t -tests based on equation (15.36) show that there are no statistically significant differences between the Fixed Effects and Random Effects estimates for `small` ($t = 1.1460$), `aide` ($t = 0.1284$), `white_asian` ($t = 1.2181$), and

Table 5: Individual Variable Hausman t -tests (Equation 15.36)

Variable	$\hat{\beta}_{FE}$	$\hat{\beta}_{RE}$	$se(\hat{\beta}_{FE})$	$se(\hat{\beta}_{RE})$	t -statistic	p -value
small	6.4902	6.4587	0.9130	0.9125	1.1460	0.2518
aide	0.9961	0.9921	0.8817	0.8812	0.1284	0.8978
tchexper	0.2856	0.3027	0.0708	0.0703	$-1.9377^\dagger$	0.0527
boy	-5.4559	-5.5121	0.7276	0.7276	$—^a$	$—$
white_asian	8.0280	7.3505	1.5357	1.4314	1.2181	0.2232
freelunch	-14.5936	-14.5843	0.8800	0.8747	-0.0956	0.9239

Note: † 顯著水準達 10% ($p < 0.1$)。

^a 因 boy 在兩模型中的標準誤完全相同 ($se = 0.7276$)，導致公式 (15.36) 的分母為零或負值，故無法計算有限樣本下的 t 統計量。

freelunch ($t = -0.0956$), since all absolute t -statistics fall well below the critical value of 1.96. The only variable that shows a noticeable difference is tchexper ($t = -1.9377$, $p = 0.0527$), which is marginally significant at the 10% level, implying that teacher experience may not be completely randomly distributed across schools due to a minor sorting effect where more experienced teachers cluster in certain types of schools. When applying this formula to the coefficient on boy, it produces an undefined result (NaN) because the standard errors under both models are exactly identical (0.7276), reducing the denominator $\sqrt{se(\hat{\beta}_{FE})^2 - se(\hat{\beta}_{RE})^2}$ to zero. This algebraic limitation occurs because student gender is naturally balanced and completely orthogonal to any unobserved school-level characteristics ($Cov(\text{boy}_{ij}, \alpha_i) = 0$), meaning that both the within-school and random-effects structures extract the exact same variation and yield identical estimation efficiency.

(f)

Table 6: Mundlak Joint Wald Test Results

Hypothesis	Df	Residual Df	F -Statistic
$H_0 : \bar{X}_{\text{small}} = \bar{X}_{\text{aide}} = \bar{X}_{\text{tchexper}} = \bar{X}_{\text{boy}} = \bar{X}_{\text{white_asian}} = \bar{X}_{\text{freelunch}} = 0$	6	5,695	2.2537

Note: ** indicates statistical significance at the 5% level ($p < 0.05$).

Variance-Covariance Matrix: School-level clustered robust covariance matrix (vcovCL) applied.

To conduct the Mundlak test, school-specific averages for all time-invariant and student-level characteristics are included in the regression, and a joint Wald test is performed on these group means using school-clustered robust standard errors. Under the null hypothesis that the unobserved school heterogeneity is uncorrelated with the explanatory variables, the coefficients on all school-average variables should be jointly equal to zero. The joint test results in an F -statistic of 2.2537 with a p -value of 0.0356. Since the p -value is less than the 5% significance level, we reject the null hypothesis, concluding that there is statistically significant correlation between the explanatory variables and the unobserved school-level heterogeneity. This outcome confirms that the Random Effects assumptions are violated, and therefore, the Fixed Effects specification must be maintained to ensure consistency.